This Week's Plan

Key idea

- Last week : the Nash equilibrium is unsatisfactory for sequential games; we introduced **Subgame-perfect equilibria**, found by **backward induction**.
- This week : more sequential games, with a common theme : **backward induction can also lead to unrealistic and unintuitive solutions**.
- New topic : games of incomplete information, also called **Bayesian games**.

The Pirate Game

Example

Rules – 5 pirates A, B, C, D, E share 100 coins.

- Captain A proposes how to split the 100 coins. If at least **half** (3 in this case, including himself) the pirates vote yes, the vote passes.
- If the vote does not pass, A is killed and B becomes captain, and makes a new proposal, that needs to be voted by at least half (2 in this case) of the remaining pirates, otherwise B is killed and C becomes captain…
- A pirate needs to be **strictly better off** with the proposed plan than in the continuation of the game to vote yes.

What's the Subgame-Perfect equilibrium? Will captain Amaro be thrown in the ocean? If not, how many of the 100 coins will he be able to keep?

Pirates left at each time

t=1	t=2	t=3	t=4	t=5
A	B	C	D	E
B	C	D	E	
C	D	E		
D	E			
E				

Key formulas

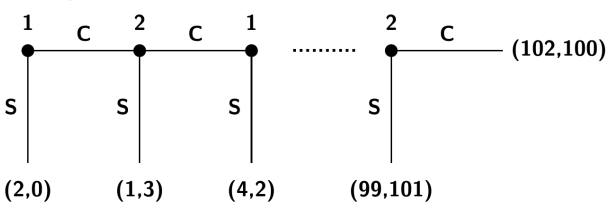
Backward induction (coins of A, B, C, D, E; "–" = already killed)

- **t=4** (D, E) : D's own vote is half of 2 : D keeps everything. (–, –, –, 100, 0)
- **t=3** (C, D, E) : C needs 2 votes. E gets 0 at t=4, so 1 coin makes him strictly better off. (–, –, 99, 0, 1)
- **t=2** (B, C, D, E) : B needs 2 votes. D gets 0 at t=3 : 1 coin buys his vote. (–, 99, 0, 1, 0)
- **t=1** (A, …, E) : A needs 3 votes. C and E get 0 at t=2 : 1 coin each. (98, 0, 1, 0, 1)

Key idea

SPE : captain A proposes $(98, 0, 1, 0, 1)$; A, C and E vote yes. He is not thrown in the ocean, and keeps **98 coins**.

The Centipede Game



Example

- Players : 1 and 2, moving in turn : Continue (C) or Stop (S).
- In each period, the total payout (sum of the payouts of Player 1 and Player 2) increases by 2 units. One would think players would want to continue playing as long as possible.

Key idea

Backward induction

- At the last node, it's optimal for Player 2 to stop : (99,101) instead of (102,100).
- At each node, knowing that at the next node the other player will stop, the player who moves at that time finds it optimal to stop.
- The unique SPE involves Player 1 stopping at the first node, with payout **(2,0)** : each player quits at every node.

Lost opportunity to gain a total of up to 202 for the players! Is backward induction realistic?

Example

The Centipede game in the lab

- People asked to play this game typically do not quit in the first stages.
- Interpretation : people have a higher tendency to collaborate than predicted by the model. They understand that if they both continue, their payout will increase over time, so they take the risk to continue, which pays off if also the other player continues.
- Different from the Prisoner Dilemma : one player has an incentive to cooperate if also the other player cooperates.

The Dollar Auction Game

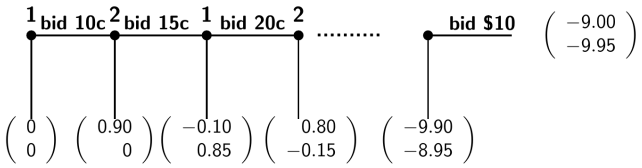
How much would you pay to get one dollar? When people play this game in lab experiments, they end up paying much more than a dollar!



Definition

DOLLAR AUCTION – FINITE-HORIZON VERSION

- Players : 1 and 2.
- Player 1 starts and can decide to bid or stay out. If she stays out, the game is over and everybody gets 0. Otherwise she bids 10 cents.
- Then at each period, the player that did not play last period can decide to quit or outbid the other player by 5 cents.
- When a player quits, the other player gets the dollar, but **both players need to pay their highest bid**.
- If no player quit before, the game ends when one player's bid reaches \$10.



Key idea

The longer the game continues, the worse off is the total payout.

Backward induction (from the end) :

- **T=199** (last period) : Player 1 is sure to get the dollar if he bids, so by bidding he gains \$0.90 for sure (relative to not bidding).
- **T=198** : Player 2 quits for sure : if he bids, in the last period Player 1 will bid again for sure and get the dollar. By bidding, Player 2 loses \$0.10 for sure (relative to not bidding).
- **T=197** : Player 1 bids for sure, because, since Player 2 quits for sure at T=198, by bidding at T=197 Player 1 gains \$0.90 for sure.
- **T=196** : Player 2 quits for sure, because at T=197 Player 1 bids for sure → by quitting, Player 2 saves \$0.10 relative to bidding.
- … Same reasoning until the beginning of the game.

Key idea

SPE : Player 1 bids in every one of his decision nodes, Player 2 quits in every one of his decision nodes.
Player 1 bids \$0.10 at T=1, Player 2 quits at T=2 : the payout is **(0.90, 0)**.

New version : the game ends when a bid *exceeds* \$10. Then the last player (if nobody quits before) would be Player 2, when he bids \$10.05.

Key idea

- From the end : Player 2 will continue in the last period (T=200), so Player 1 will quit at T=199, so Player 2 will continue at T=198…
- SPE : Player 2, with the advantage of being the last if nobody quits before, will always continue, and Player 1 will always quit.
- But since Player 1 starts, he will not even enter the game : the payout is **(0,0)**.

Dollar Auction : Infinite-Horizon Version

In the finite game, one player is sure to be the one that bids in the "final period" (when bids reach or exceed the \$10 threshold), so he is sure to win if nobody quits. The other player is sure to lose : he stays out from the beginning.
What if there is no final time and the auction continues until somebody drops out? (Player 1 moves in every odd period, Player 2 in every even period.)

Key idea

If nobody quits between a certain time t and $t+2$, **nothing has changed between** t **and** $t+2$, except the fact that both players have already committed to pay 10c more than at period t.
But this is a **sunk cost** that should not affect the continuation of the game!

Stationary strategies : each player quits or continues with the same probability at each of her decision nodes. At a node where Player i has already committed to pay x :

Key formulas

If she continues :

$$P_i(x) = \begin{cases} -0.1 + P_i(x) & \text{if Player } -i \text{ continues} \\ -x + 0.9 & \text{if Player } -i \text{ quits} \end{cases}$$

If she quits :

$$P_i(x) = -x$$

- If Player $-i$ always continues, $P_i(x) = -\infty \rightarrow$ optimal to quit.
- If Player $-i$ always quits, optimal to continue.

Mixed strategies : for one player to randomize, she needs to be indifferent between quit and continue, which is only possible if the other player also randomizes. If Player $-i$ continues with probability q :

Key formulas

$$P_i(x; C) = q(-0.1 + P_i(x)) + (1-q)(0.9 - x)$$

$$P_i(x; Q) = -x$$

Player i is indifferent if the two payouts are the same :

$$q = 0.9$$

Key idea

3 strategy profiles that are NE :
- $(Quit, Continue)$
- $(Continue, Quit)$
- $((0.9C, 0.1Q), (0.9C, 0.1Q))$

Only one of these three, **(Continue, Quit)**, survives if the game has a finite horizon – even if the final time is after, say, 1 million periods, and no player would ever want to get there! Problematic feature of backward induction…

Example

In lab experiments : people continue to bid without problem until the highest bid reaches $1. Afterwards, they start to feel uncomfortable but often continue to outbid each other for a while. Typically the dollar ends up being sold for $3 (plus almost $3 paid by the second bidder).
The concept of **sunk cost** is difficult to digest for non-experienced players.

Games of Incomplete Information (Bayesian Games)

So far, everything about the game was **common knowledge** : who are the players, what are their preferences, what are the payouts. But players might have private information about their own payout, their "type" or preferences, etc.

Definition

BAYESIAN GAME (simultaneous-move game of incomplete information)
Defined by :

- Players
- Action set for each player
- Possible types θ_i for each player i and a probability distribution over player i types $p(\theta_i)$.

A pure strategy is a plan of action **for each type of player**.

Definition

BAYESIAN NASH EQUILIBRIUM (BNE)
A strategy profile $(s_1(\theta_1), s_2(\theta_2), \dots, s_N(\theta_N))$ is a BNE of a Bayesian game with N players iff, for each player i, her **expected payout** using strategy i is higher than or equal to her expected payout using any of the other available strategies, keeping fixed the strategies of the other players, which **depend on their types** :

$$E[\mathcal{P}_i(s_i(\theta_i), s_{-i}(\theta_{-i})|\theta_i)] \geq E[\mathcal{P}_i(\tilde{s}_i(\theta_i), s_{-i}(\theta_{-i})|\theta_i)]$$

The expected value is taken over all possible types of the other players.

Example : Public Good and Free Riding

Example

- Two players can work (W) or not work (N) to produce a public good, that would give utility 1 to both players.
- If at least one player works, the public good is available for both. If nobody works, it is not available.
- Working has a cost c_1 for Player 1 and c_2 for Player 2.

$1\backslash 2$	W	N
W	$(1-c_1, 1-c_2)$	$(1-c_1, 1)$
N	$(1, 1-c_2)$	$(0,0)$

- $c_1 = 0.5$, known to everybody.
- c_2 is known to Player 2 but not to Player 1 : it can be 0.2 with probability p and 1.2 with probability $1-p$.
- **The value of p is common knowledge.**

Equivalently, Player 2 can be of two **types** : **Efficient** (E, cost of working $c_2 = 0.2$, probability p) or **Inefficient** (I, $c_2 = 1.2$, probability $1-p$).

prob p

$1\backslash$ 2E	W	N
W	(0.5,0.8)	(0.5,1)
N	(1,0.8)	(0,0)

prob $(1-p)$

$1\backslash$ 2I	W	N
W	(0.5,-0.2)	(0.5,1)
N	(1,-0.2)	(0,0)

Key idea

'Work' is a **dominated** strategy for the Inefficient type of Player 2.

Scenario 1 : Efficient Player 2 works, Inefficient Player 2 does not work. Player 1's best reaction ?

Key formulas

- Player 1 works : $\mathcal{P}_1 = 0.5$ for sure.
- Player 1 does not work : payout 1 if Player 2 works (E type), 0 otherwise (I type) :

$$\mathcal{P}_1 = p \times 1 + (1-p) \times 0 = p$$

$\Rightarrow$ Player 1 works if $p < 0.5$, and does not work if $p > 0.5$ (indifferent if $p = 0.5$).

- For $p > 0.5$ Player 1 does not work (for $p = 0.5$ he may not work). Knowing this, Player 2 works when he is of type E : the starting point was correct.
  $\Rightarrow$ **If $p \geq 0.5$, one BNE is** (N, WN) : N for Player 1, W for Player 2 of type E, N for Player 2 of type I.
- If $p < 0.5$, Player 1 works; but then Player 2 will not work even if he is of efficient type...Hence for $p < 0.5$ this scenario is not a BNE.

Scenario 2 : Player 2 never works. Player 1 gets $1 - c_1 = 0.5$ if he works and 0 if he does not $\Rightarrow$ regardless of p, Player 1 works; and when Player 1 works, Player 2 prefers not to work (1 > 0.8 and 1 > -0.2). **BNE :** (W, NN).

Key idea

- If $p < 0.5$: unique pure-strategy BNE (W, NN).
- If $p \geq 0.5$: two pure-strategy BNE (W, NN) and (N, WN).

Harsanyi's Idea

Key idea

Imagine that "Efficient" and "Inefficient" are two states of the same person (e.g. depending on whether the player slept well the previous night).
Nature is a third player who moves first and draws the type of Player 2 (prob p Efficient, prob $(1-p)$ Inefficient). Player 2 needs to find a strategy in the two possible states (e.g. "Work if efficient, Not Work if Inefficient") to maximize his **ex-ante expected payout**.

Definition

HARSANYI REPRESENTATION
Collapses all the payout tables contingent on the players' types into **one table**. The columns contain the strategy of Player 2 conditional on his type : e.g. 'WN' means "Work if efficient, Not Work if Inefficient".

$1\backslash 2$	WN	NN	NW	WW
W	(0.5, 1-0.2p)	(0.5, 1)	(0.5, 1.2p-0.2)	(0.5, p-0.2)
N	(p, 0.8p)	(0,0)	(1-p, -0.2(1-p))	(1, p-0.2)

Key formulas

Payout for the (A_1, A_2A_2') strategy $= p \times$ payout of (A_1, A_2) in the table of the Efficient Player 2 $+ (1-p)$ $\times$ payout of (A_1, A_2') in the table of the Inefficient Player 2 (A_1, A_2, A_2' can be W or N).

After elimination of the dominated strategy W for the Inefficient Player 2 :

$1\backslash 2$	WN	NN
W	(0.5, 1-0.2p)	(0.5, 1)
N	(p, 0.8p)	(0,0)

Key idea

Find the BNE the standard way on the big table : we find the same pure-strategy equilibria as before : (W, NN) if $p < 0.5$, (W, NN) and (N, WN) for $p > 0.5$.

Weekly Summary

Summary

- **Pirate game** : solved by backward induction, captain A keeps 98 coins.
- **Centipede game** : the unique SPE stops at the first node, while people in the lab continue.
- **Dollar auction** : in finite horizon the SPE has the dollar sold for 10c (or no entry); in infinite horizon three NE; sunk costs should not matter.
- Backward induction can lead to unrealistic and unintuitive solutions.
- **Bayesian games** : types, a strategy for each type, Bayesian Nash equilibrium.
- **Harsanyi** : Nature draws the types; one table with type-contingent strategies.